

0 3

A student prepared a stained squash of cells from the root tips of garlic to calculate a mitotic index. He:

1. cut the end 5 mm from 10 garlic roots
2. placed the root tips into a Petri dish containing 5 cm³ of hydrochloric acid for 12 minutes
3. rinsed the root tips in distilled water
4. placed one of the root tips on a microscope slide and added toluidine blue stain
5. placed a coverslip onto the microscope slide, and gently pressed the coverslip downwards on the root tip
6. observed the root tip using an optical microscope.

0 3**. 1**

Suggest why the student soaked the root tips in hydrochloric acid in step 2.

[2 marks]

0 3**. 2**

Pressing the coverslip downwards enabled the student to observe the stages of mitosis clearly.

Explain why.

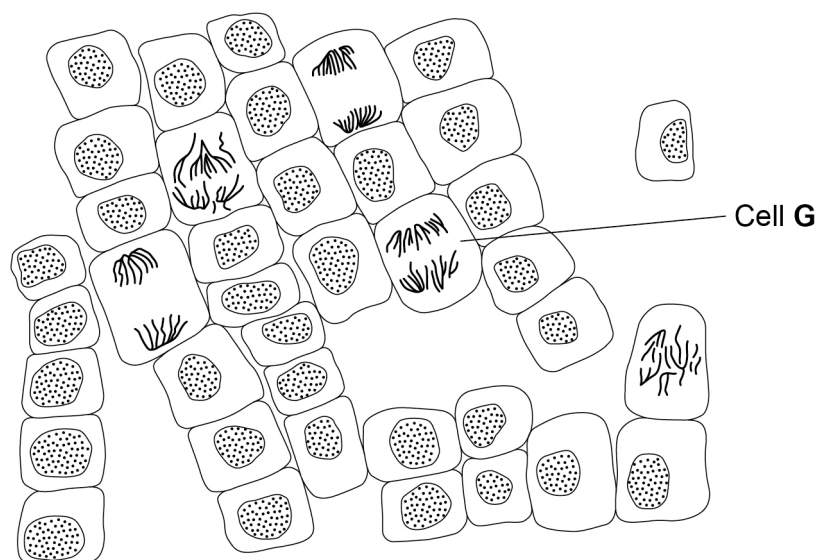
[2 marks]

Question 3 continues on the next page

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Figure 6 shows the student's drawing of one field of view.

Figure 6



0 3 . 3

Name the stage of mitosis shown in cell **G**. Explain the appearance of this cell.

[2 marks]

Stage of mitosis _____

Explanation _____

0 3 . 4

Use **Figure 6** to calculate a mitotic index for the cells in this field of view.

[1 mark]

Mitotic index _____

0 3 . 5

Other students in the class followed the same method, but calculated different mitotic indices.

Apart from student errors, suggest **two** explanations why.

[2 marks]

1 _____

2 _____

9

Turn over for the next question

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